

PYTHON SETS

> WHAT IS A SET??

Python set is a collection of unique elements that are unordered. Items in the set do not have index positions. We use curly brackets '{ }' to define the set.

The type function will return <class 'set'>.

Example:

```
first_set = {1, 2, 3, 4, 4, 4, 4, 5, 5, 'string', 24.8}  
print(first_set)
```

Output:

```
{1, 2, 3, 4, 5, 'string', 24.8}
```

> CHARACTERISTICS OF A SET

Let us look at the characteristics of the Set one by one:

CHARACTERISTIC	MEANING
1> Unchangeable	We can not change the items stored within the set. We can add or delete items to set.
2> No Duplicates	Duplicate items are not allowed in a set.
3> Unordered	Items in a set do not have defined order.
4> Heterogeneous	Sets can store elements of various data types, but only immutable values.

Fig: Set Characteristics

> SETS ARE UNCHANGEABLE

Items in a set do not have index positions, so there is no point in changing any value in the set using its index position.

Example:

```
numbers_set = {1, 2, 3, 4, 5, 8, 9}
numbers_set[1] = 7
```

Output:

`TypeError: 'set' objects does not support item assignment.`

> SETS ARE UNORDERED

Items in the set do not follow any specific order. Every time, the content of the set may appear in different order.

The order in which we assigned the items and the order in which we get the items can be different.

Example:

```
numbers-set = {91, 36, 26, 78, 14, 89, 67}
print(numbers-set)
```

Output:

```
{67, 36, 78, 14, 89, 26, 91}
```

> CREATING AN EMPTY SET

To create an empty set, we use the `set()` function.

Example:

```
emptyset = set()
print(type(emptyset))
```

Output:

```
<class 'set'>
```

> LOOPING SET

We can loop through elements of the set using for loop.

Example:

```
my-set = {'a', 'b', 'c', 'd'}
for i in my-set:
    print(i)
```

Output:

```
a
d
c
b
```

> THE len() function

The len() function can be used to check the number of items in the set.

Example:

```
my_set1 = {1, 2, 3, 4}
print(len(my_set1))
```

Output:

4

> MEMBERSHIP TEST

Example:

```
my_set1 = {1, 2, 3, 4}
print(1 in my_set1)
print(1 not in my_set1)
```

Output:

True
False

PUACP